

Medical Tourism and Clinic Clustering

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Health Inequality and Medical Tourism

- ▶ Medical tourism consists of temporarily visiting a country with the main objective of attaining medical care.
- ▶ Key Reasons
 - ▶ Lower Costs
 - ▶ Shorter Wait Times
 - ▶ Better Healthcare
 - ▶ Legal differences
- ▶ Medical tourism has contributed to the economic growth and specialization of certain cities, but has also brought unintended consequences

- ▶ Los Algodones, Baja California—also known as “Molar City.” [Clement et al., 2024]
 - ▶ The city has an exceptionally high concentration of dental care clinics serving foreign patients.
 - ▶ Local marginalized populations may be adversely affected through a crowding-out effect, as resources and services cater primarily to medical tourists.
- ▶ Insulin and Fertility Treatment
 - ▶ Procedures and medicines that are expensive in the U.S. can cost a fraction in Mexico.
 - ▶ Tijuana, Monterey, and Juarez are the biggest cities for this type of medical tourism.
 - ▶ IVF (In vitro fertilization) which can be around \$18,000 in the us can cost \$7,500
 - ▶ Insulin can be purchased for reduced prices like for example \$16 vs \$110 for an insulin pen

Question and Data Source

Question

- ▶ Is there any evidence of clustering of medical units on U.S./Mexico border towns?

Data

- ▶ The data was collected for 2025 September.
- ▶ Data on medical units comes from the Mexican Secretary of Health; it is the universe of codes given to all medical units in Mexico, both private and public. The data contains coordinates for each establishment, as well as several indicators for rural/urban and other types of care.
- ▶ Data on population per county and the shape file for all of Mexico with counties comes from the Mexican National Institute for Statistics and Geography.

Summary Statistics

Table 1: Summary Statistics for Population and Clinic Measures

Variable	Min	Q1	Median	Mean	Q3	Max
Population	349	3,098	9,230	89,487	44,852	2,151,740
Number of Clinics	1.00	5.00	10.50	33.32	28.00	635.00
Expected Clinics	0.13	1.15	3.44	33.32	16.70	801.11
Clinics per 10,000	0.75	4.19	9.87	14.05	19.19	80.65

Exploratory Analysis

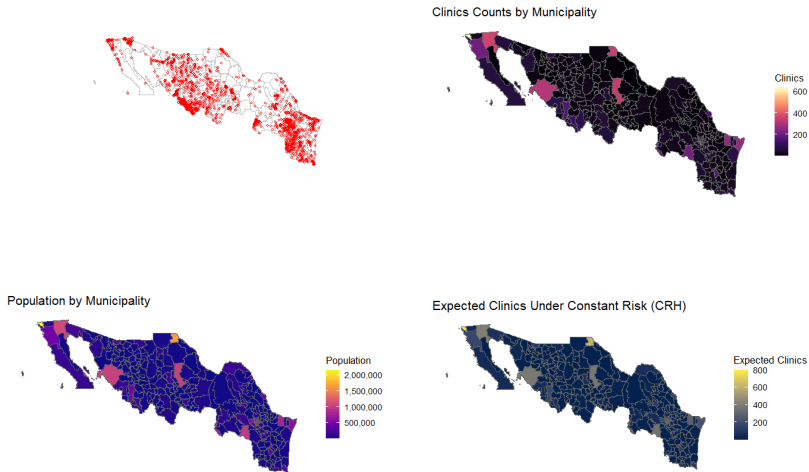


Figure 1: Clinic Distribution, Population, and Expected Clinics Under Constant Risk

Besag-Newell Test

► Local Test

- Ho The most compact window regarding population with at least c^* hospitals on the border U.S. Mexico is not significantly more compact than what is expected under the constant risk hypothesis
- Ha The most compact window regarding population with at least c^* hospitals on the border U.S. Mexico is significantly more compact than what is expected under the constant risk hypothesis

Besag–Newell Method

Threshold c^*	Cases	Population	p-value
10.5	11	1,364	1.98×10^{-10}
28	28	5,091	3.15×10^{-23}
33.32	39	7,806	1.34×10^{-23}
68.3	71	16,118	9.06×10^{-47}

Table 2: Besag–Newell cluster results for Northern Mexico across multiple c^* thresholds.

- The way I picked the c^* was by getting the mean and the following quantiles 50th, 75th, 90th,

Besag–Newell Method Maps

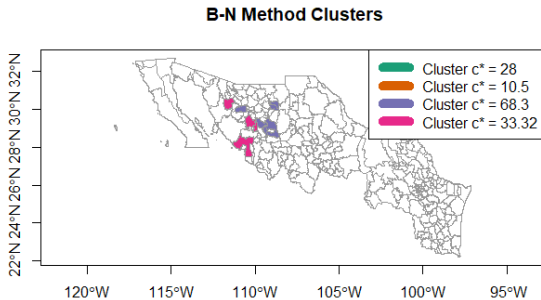


Figure 2: Besag-Newell Clusters

CEEP Test

- ▶ Local Test

- Ho There is no window with n^* persons that has significantly more clinics in the U.S. Mexico border states than what is expected under the CRH
- Ha There is at least one window with n^* persons that has significantly more clinics in the U.S. Mexico border states than what is expected under the CRH

CEEP Test Maps

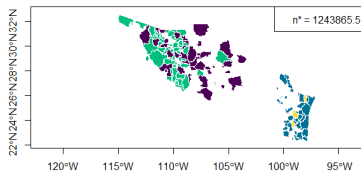
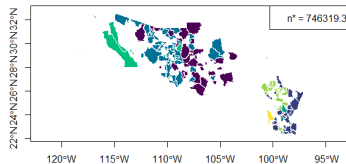
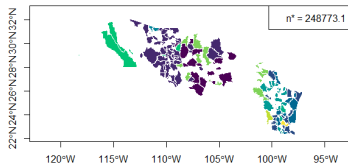


Figure 3: CEEP Clusters at Multiple Population Scales

CEEP Test Discussion

- ▶ The way I picked the n^* was by getting clusters that contain 1% 3% and 5% of the total population. Since this helps us look at very small local clusters to larger regional concentrations of populations.
- ▶ We see that at these n^* values the test stays robust and consistent apart from some major metropolitan areas like Tijuana.

Scan Test (Kulldorff)

- ▶ Local Test

Ho The most likely cluster of clinics is consistent with what is expected under the constant risk hypothesis, regarding the local rate of clinics compared outside of the cluster

Ha The most likely cluster of clinics is consistent with what is more extreme than what is expected under the constant risk hypothesis, regarding the local rate of clinics compared outside of the cluster

Scan Test (Kulldorff) Map

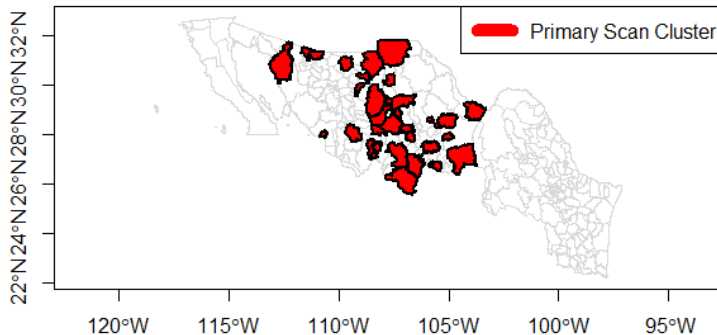


Figure 4: Scan Test

Scan Test (Kulldorff) Municipalities

Border Municipalities	Nearby Municipalities
Manuel Benavides	Ascensión
Matamoros	Empalme
Nogales	Caborca
Janos	Sáric
	Bacerac
	Bacadéhuachi

Table 3: Primary Kulldorff scan cluster municipalities located on or near the U.S.–Mexico border.

- ▶ As we see, there are only around 4 border municipalities and around 10 total municipalities most of which are very small in size.

Scan Test (Kulldorff) Discussion

Adjusted Moran's I

► Global test

H₀ Our measure of global spatial autocorrelation is zero

H_a Our measure of global spatial autocorrelation is not zero

Surface	Moran's I	p-value
Observed Clinics	0.16714	0.002
Expected Clinics (CRH)	0.16710	0.001

Table 4: Monte Carlo simulation results of Moran's I for observed and expected clinic counts in Northern Mexico (999 simulations).

Adjusted Moran's I Discussion

- ▶ Eventhough this is the adjusted Moran's I using the expected amount of clinics. When run with "number of clinics", the difference in the statistic was insignificant.

Conclusion

References I



Clement, V., Lemus, S., and Newman, E. (2024).

In search of health: medical tourism at the us–mexico border/lands.

Journal of Tourism and Cultural Change, 22(1):118–136.